

Embedded AI – Assignment 6

EMBD X424

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https://github.com/electrickitty2/embedx424_assignment_6

Assignment: Building a Tiny Transformer That Learns Addition

Overview:

- In this assignment you will implement a minimal GPT-style Transformer inspired by:
- The Tiny Transformer walkthrough notebook
- Andrej Karpathy's nanoGPT / miniGPT style training loops
- The projects/adder sequence modeling idea

Learning Goals:

- By the end of this assignment you should be able to:
 - Build a tokenizer and character vocabulary
 - Generate autoregressive training data
 - Implement causal self-attention from scratch
 - Assemble a Transformer block
 - Train a tiny GPT-like language model
 - Generate predictions autoregressively
 - Evaluate model quality using real mathematical checks

Submission Requirements:

- 1) Completed notebook
- 2) Final autograder score
- 3) Screenshot of generated examples
- 4) Short explanation:
 - 1) What was hardest?
 - 2) What debugging strategy worked best?
 - 3) What surprised you about Transformers?

Completed Notebook:

- Colab link:
 - https://colab.research.google.com/drive/1sUPab9HE1jxBa_EdS1ODXGI_99yISvK-?usp=sharing
- Github link:
 - https://github.com/electrickitty2/embedx424_assignment_6/blob/main/assignment_6.ipynb

Final Autograder Score

```
=====
AUTOGRADER RESULTS
=====
```

```
[PASS] Tokenizer encode/decode works
[PASS] Batch generation works
[PASS] Attention head runs
[PASS] Model forward pass works
```

```
2-digit accuracy: 86.00%
[FAIL] Accuracy below 95%
```

Example predictions:

```
{'prompt': '54+83=', 'prediction': '137', 'expected': '137', 'correct': True}
{'prompt': '35+79=', 'prediction': '114', 'expected': '114', 'correct': True}
{'prompt': '74+41=', 'prediction': '115', 'expected': '115', 'correct': True}
{'prompt': '23+90=', 'prediction': '113', 'expected': '113', 'correct': True}
{'prompt': '24+19=', 'prediction': '43', 'expected': '43', 'correct': True}
{'prompt': '40+25=', 'prediction': '65', 'expected': '65', 'correct': True}
{'prompt': '93+60=', 'prediction': '153', 'expected': '153', 'correct': True}
{'prompt': '60+82=', 'prediction': '142', 'expected': '142', 'correct': True}
{'prompt': '37+99=', 'prediction': '136', 'expected': '136', 'correct': True}
{'prompt': '23+20=', 'prediction': '43', 'expected': '43', 'correct': True}
```

```
=====
FINAL SCORE: 70/100
=====
```

Example:

```
[*]: #  
# Examples  
#  
while( 1 ):  
    prompt = input("Enter equation:")  
    result = generate( model, prompt.strip(), max_new_tokens=10, temperature=0.1)  
    print( "Model output:", result)
```

Enter equation: 10+10=
Model output: 10+10=20
95+63=1
Enter equation: 10+10=
Model output: 10+10=20
97+96=1
Enter equation: 10+10=
Model output: 10+10=20
84+46=1
Enter equation: 84+46=
Model output: 84+46=130
71+78=

Enter equation:

Short explanation:

- What was hardest?
 - Frankly I struggled with transformers. Even during the lecture I was struggling.
 - It reminded me of my crypto class. I understood everything until public key exchange. Then my mind was blown and I still don't understand this.
-

Short explanation:

- What debugging strategy worked best?
 - I was very reliant on Claude and Gemini. I would ask Claude to explain and suggest next steps and I would type in one character and Gemini would finish the line for me.

Obviously this is a well published problem.

This got me to 92% accuracy, but this plateaued

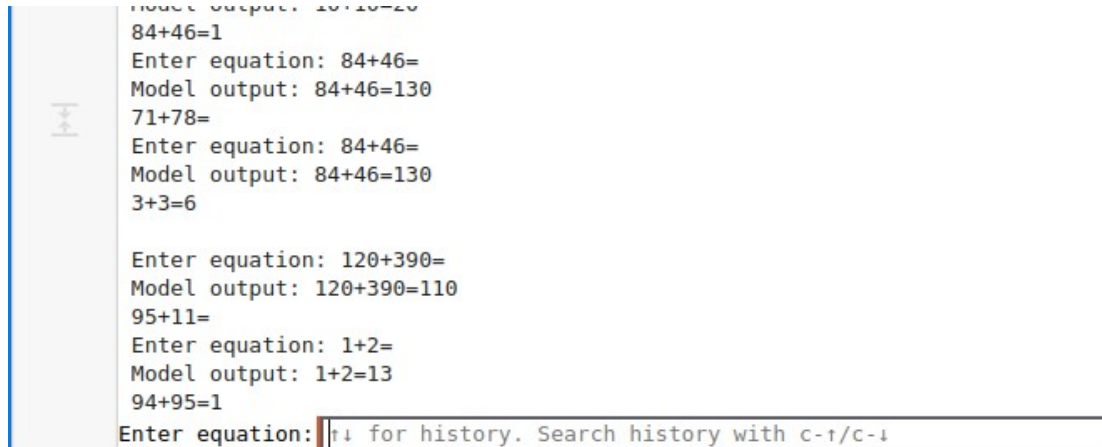
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Short explanation:

- What surprised you about Transformers?
 - The explanation in class seemed simple, and I thought I understood. Until it was clear I did not.
 - Then the words and explanations did not make sense to me at all.
 - No matter how hard I tried to get above 92% accuracy, nothing impacted the result. Claude code was telling me I was on the wrong track the whole time.

Short explanation:

- What surprised you about Transformers (cont)?
 - I have no idea why the example, emits additional equations that are wrong. I would expect it to stop once the answer was known.



```
model output: 84+46=1
84+46=1
Enter equation: 84+46=
Model output: 84+46=130
71+78=
Enter equation: 84+46=
Model output: 84+46=130
3+3=6

Enter equation: 120+390=
Model output: 120+390=110
95+11=
Enter equation: 1+2=
Model output: 1+2=13
94+95=1
Enter equation: 
```